Andrea Galenda

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github.com/andreagalenda | andreagalenda.github.io/Projects

PROFILE

Mechanical Engineering graduate from Politecnico di Milano (MSc, 110/110 cum laude) and former Visiting Student Researcher at NASA Jet Propulsion Laboratory, specializing in multi-physics dynamics, robotics, control and state estimation. Experienced in modeling complex mechanical systems for space applications and highly motivated to pursue a career in field robotics within international engineering environments.

EDUCATION

Politecnico di Milano Sept. 2023 – Oct. 2026
MSc. in Mechanical Engineering (Track: Mechatronics and Robotics) – ***110/110 cum laude*** Milan, Italy

- **Thesis**: *Modelling, Dynamics and In-Flight Leakage Detection for a Variable Altitude Venus Aerobot* (NASA JPL)
- **Relevant coursework**: Optimal Control and State Estimation, Advanced Dynamics of Mechanical Systems, Advanced Machine Design, Actuating Devices, Signal Processing

Polytech Paris-Saclay Sept. 2024 – Feb. 2025
Erasmus+ Program, Electronics and Robotics Engineering Paris, France

Politecnico di Milano Sept. 2020 – Sept. 2023
BSc. in Mechanical Engineering Milan, Italy

EXPERIENCE

NASA Jet Propulsion Laboratory Dec. 2025 – Aug. 2026
Visiting Student Researcher - Aerial Robotic Systems (Section 358M) Pasadena (CA), United States

- Developed a 6-DOF flight dynamics and thermodynamics simulator in MATLAB for a variable-altitude Venus Aerobot, coupling flight dynamics, envelope shape, a six-node thermal network and gas transport in a fifteen-state system
- Engineered a Physics-Informed Neural Network (PINN) surrogate to rapidly predict the dynamic shape of the Balloon-in-Balloon configuration
- Implemented an Unscented Kalman Filter (UKF) on existing avionics (barometer, IMU, thermistors) to estimate the specific-force deficit caused by gas leakage
- Designed a CUSUM-based change detection pipeline with constrained regression, detecting every leak from 1 to 100 g/day within about a week of onset and separating it from sulphuric acid deposition

SELECTED PROJECTS

SayCan Framework for Pick & Place

- Built a SayCan framework for language-guided autonomous pick-and-place, combining LLMs with vision-language detection (ViLD) and a fine-tuned CLIPort grasping policy, with geometric checks to verify task success

Mechanism Design and Robotics

- Designed and 3D printed an electrically actuated end-effector to operate a modular helmet mechanism; implemented robot path planning in Python with task verification via camera and torque feedback

Bayesian Optimization for Robot Controller Tuning

- Automated PID tuning of a multi-joint robot with two-phase Bayesian Optimization (LCB exploration, EI+ exploitation)

Deep Learning Reduced Order Models for Structural Simulation

- Trained POD-NN and DL-ROM surrogates to replace FEniCS elastodynamics solves (real-time, under 2.5% L2 error)

SKILLS & INTERESTS

Simulation and Physics: MATLAB/Simulink, Python, Flight physics modeling, System dynamics, ROS, Gazebo
Estimation and Data Analysis: Kalman filtering (UKF), Change detection, Signal Processing, C++
CAD/CAE: SolidWorks, CATIA, Autodesk Inventor, Onshape, Solid Edge, FEniCS
Languages: English (C1), French (B2), Italian (Native)
Sports: Competitive rugby player, Serie A Italy and Fédérale 3 France (2016 – 2025)